MIDSEM PROJECT PRESENTATION

GenoVista: Disease causing Genomic Variant Interpretation using AI-ML

Team Details

Group No. : 127

Member 1: Tanmay Bansal, 2K22/IT/170

Member 2: Shruti Singh, 2K22/IT/158

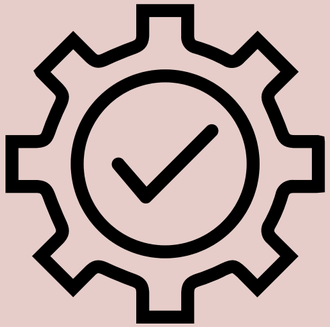
Member 3: Priyanshu Bajaj, 2K22/IT/127

Supervisor:

Dr. Bindu Verma

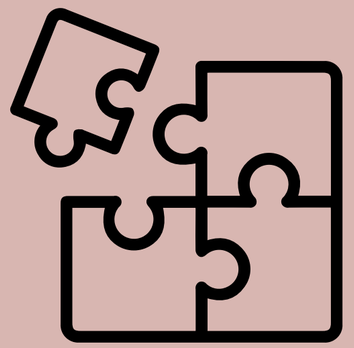
PROGRESS SO FAR

Percentage Complete: 25% - The primary focus has been on developing and validating the core analytical backbone (Back-end/API) of the application



Infrastructure Setup

Successfully configured a serverless Python environment using Modal. This handles the environment build, dependency management (including CUDA/PyTorch), and integration of the EVO 2 model library.



Model & API Integration

Integrated the EVO 2 (7B parameter) model checkpoint with persistent caching (Hugging Face Cache). The core analysis endpoint is fully implemented, tested, and live on Modal.

Core Logic

Implemented analysis logic: Fetches 8,192-base sequence (UCSC API), calculates Delta Score (EVO 2 model call), and translates result to a "Likely Pathogenic/Benign" prediction using the calculated ROC threshold.

CODE BASE LINK

LITERATURE REVIEW:

1. Evo 2 (2025)

2. Machine Learning for Variant Classification (2022–2025)

3. Advances in Clinical Databases: ClinVar (2022–2025)

4. Comprehensive Variant Analysis Platforms (2024–2025)

TASKS LEFT TO BE DONE:

The remaining work is focused on building the user interface, full feature integration, and project scaling/documentation.

Front-end Structure & UI

- Implement the core Next.js/React application structure, navigation, and build all primary user components (Gene Viewer, Main Dashboard, Comparison Modal).

External Data Integration

- Integrate all auxiliary APIs (UCSC, ClinVar, NCBI) to handle fetching genomic data (assemblies, chromosomes) and displaying known clinical variants.

Visualization & Core Features

- Implement the interactive Sequence Viewer (slider, coloring, mutation selection) and link the UI forms and tables to the deployed Modal API endpoint.

Quality Assurance (QA) and Threshold Research

- Conduct rigorous end-to-end testing, bug fixing across genome assemblies (HG19, HG38), and performance optimization (reducing API latency).

Documentation & Publication

- Finalize the B. TECH Report and all technical documentation. Refine data analysis and draft the Research Paper for publication submission.

EXPECTED TIMELINE OF PROJECT COMPLETION:

AUG - OCT

1

FOUNDATION & BACK-END CORE

Finalize Python analysis logic, model integration, and cloud API deployment (Modal). Begin Front-end structure definitio

NOV

2

FRONT-END FRAMEWORK IMPLEMENTATION

Implement core Next.js/React structure. Complete Genome Assembly and Chromosome selection features

DEC

3

VISUALIZATION & CORE DATA INTEGRATION

Complete the interactive Sequence Viewer, known clinical variants table (ClinVar), and coloring utilities

JAN - FEB

4

INTEGRATION & QA & THRESHOLD RESEARCH

Integrate Front-end with the live Modal API. Complete the analysis forms and comparison UI. Conduct rigorous end-to-end testing and optimization

MAR

5

FINAL SUBMISSION AND PUBLICATION

Deliver final code base and B. Tech Report. Prepare and submit Research Paper draft

Thank You